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Answer

Okay, I'm ready to help you crack JEE! Let's tackle this question on equivalence relations. It's a good one to test your understanding of sets and relations.

Understanding the Problem

This is a **Conceptual** problem. We need to determine how many different ways we can define an equivalence relation on a set containing three elements: {1, 2, 3}. Remember, an equivalence relation must satisfy three properties:

Reflexivity: Every element is related to itself.

Symmetry: If A is related to B, then B is related to A.

Transitivity: If A is related to B, and B is related to C, then A is related to C.

Step-by-Step Solution

Here's how we can solve this systematically:

Understanding Equivalence Relations and Partitions: An equivalence relation on a set naturally divides the set into disjoint subsets called equivalence classes or partitions. Elements within the same partition are related to each other, and elements in different partitions are not.

Possible Partitions of {1, 2, 3}: We need to figure out all the possible ways to divide the set {1, 2, 3} into non-empty subsets. Here's a breakdown:

Case 1: One subset

{{1, 2, 3}} (All elements in one group)

Case 2: Two subsets

{{1}, {2, 3}}

{{2}, {1, 3}}

{{3}, {1, 2}}

Case 3: Three subsets

{{1}, {2}, {3}} (Each element in its own group)

Each Partition Represents an Equivalence Relation: Each of these partitions directly corresponds to a valid equivalence relation. Let's think about why:

Case 1: {{1, 2, 3}} This means every element is related to every other element. This satisfies reflexivity, symmetry, and transitivity.

Case 2: e.g., {{1}, {2, 3}} This means 2 is related to 3, and vice versa, but 1 is only related to itself. This also satisfies all the equivalence relation properties.

Case 3: {{1}, {2}, {3}} This means each element is only related to itself. This is also an equivalence relation (the "identity" relation).

Counting the Partitions: We identified the following partitions:

1 partition with 1 subset

3 partitions with 2 subsets

1 partition with 3 subsets

Total Number of Non-Empty Equivalence Relations: Adding up the number of partitions, we have 1 + 3 + 1 = 5. However, the question asks for *non-empty* relations. All the listed relations are non-empty.

Answer:

Therefore, the number of non-empty equivalence relations on the set {1, 2, 3} is **5**. The correct answer is (3) 5.

Alternative Method (Using Bell Numbers)

For larger sets, manually listing partitions becomes tedious. There's a concept called **Bell Numbers** that helps. The *n*th Bell number, denoted as B_n, gives the number of ways to partition a set of *n* elements.

B₃ = 5 (as we just found)

You can look up or calculate Bell numbers. The first few are:

B₀=1, B₁=1, B₂=2, B₃=5, B₄=15, and so on.

Key Takeaways and Tips for JEE Preparation

Master the Definitions: Ensure you thoroughly understand the definitions of equivalence relations (reflexivity, symmetry, transitivity) and the concept of partitions.
Practice with Examples: Work through examples of different sets and equivalence relations. This helps solidify your understanding.
Visualize: Drawing diagrams can be helpful. For example, you could represent an equivalence relation with arrows connecting related elements.
Connect Concepts: Recognize the connection between equivalence relations and partitions.
Don't Memorize, Understand: While knowing Bell numbers can be helpful for a quick solution, focus on understanding the underlying concepts.
Practice Problems: Work through similar problems from NCERT textbooks, JEE previous year questions (PYQs), and other practice resources. This will help you build confidence and speed. Look for questions that involve finding the number of equivalence relations or the number of possible partitions.
Recommended Practice Problems:
NCERT: Review the examples and exercises related to relations and functions.
PYQs: Search for problems on equivalence relations and partitions from past JEE papers (both Main and Advanced).
Reference Books: Consult your standard JEE preparation books for additional practice problems.
Motivational Note: You're doing great by tackling these problems! Keep practicing, stay curious, and don't be afraid to ask questions. Every problem you solve is a step closer to your JEE goals! I am here to guide you; let me know if you have any other questions.

Ask your queries

Type your question here... Ask Analyze

Follow-ups:

- What is equivalence relation?

- what is reflexitivy property?

Okay, here's the answer to your question about reflexivity, explained simply:
The reflexivity property, in the context of this problem (which is related to math, specifically equivalence relations), means that something is related to itself.
Think of it this way:
In a regular relationship, it means that something is related to itself, so a person is related to the person.